

add factors

March 10, 2026

1 When using Mexplode og %%Mexplodemodel the tag <add> will introduce an add factor

In this we will look into some of the behavior of this

```
[24]: %load_ext autoreload
      %autoreload 2
```

The autoreload extension is already loaded. To reload it, use:

```
%reload_ext autoreload
```

1.1 Activate the %%Mexplodemodel magic cell

This is a convinient wrapper around the Mexplode function

```
[25]: import modeljupytermagic # ti activate the %% %%Mexplodemodel magic cell
```

1.2 A simple equation

1.2.1 When the equation is already normalized

That is the left hand side is a variable

```
[26]: %%Mexplodemodel test show
      > a = b
```

```
> a = b
```

```
FRML <> A = B $
```

```
[27]: %%Mexplodemodel test show
      ><add> a = b
```

```
><add> a = b
```

```
FRML <ADD> A = (B + A_A) $
```

```
FRML <CALC_ADD_FACTOR> A_A = (A) - (B) $
```

```
[28]: %%Mexplodemodel test show
      ><add> a = log(b)
```

```
><add> a = log(b)
```

```
FRML <ADD> A = (log(B) + A_A) $
```

```
FRML <CALC_ADD_FACTOR> A_A = (A) - (log(B)) $
```

1.2.2 When the equation has to be normalized no differences on left hand side

That is the left hand side is a variable

```
[29]: %%Mexplodemodel test show  
><add> log(a) = b
```

```
><add> log(a) = b
```

```
FRML <ADD> A = EXP(A_A+ (B) ) $
```

```
FRML <CALC_ADD_FACTOR> A_A = - ((B)) +LOG(A) $
```

```
[30]: %%Mexplodemodel test show  
><add> log(a) = diff(b)
```

```
><add> log(a) = diff(b)
```

```
FRML <ADD> A = EXP(A_A+ (((B)-(B(-1)))) ) $
```

```
FRML <CALC_ADD_FACTOR> A_A = - (((B)-(B(-1))))) +LOG(A) $
```

```
[31]: %%Mexplodemodel test show  
><add> log(a)+d+e = b
```

```
><add> log(a)+d+e = b
```

```
FRML <ADD> A = EXP(A_A-D-E+ (B) ) $
```

```
FRML <CALC_ADD_FACTOR> A_A = D+E- ((B)) +LOG(A) $
```

1.2.3 When the equation has to be normalized have differences on left hand side

That is the left hand side is a variable

```
[32]: %%Mexplodemodel test show  
><add> diff(a) = b
```

```
><add> diff(a) = b
```

```
FRML <ADD> A = A_A+ (B) +A(-1) $
```

```
FRML <CALC_ADD_FACTOR> A_A = A- ((B)) -A(-1) $
```

```
[33]: %%Mexplodemodel test show  
><add> dlog(a) = b
```

```
><add> dlog(a) = b
```

```
FRML <ADD> A = A(-1)*EXP(A_A+ (B) ) $
```

```
FRML <CALC_ADD_FACTOR> A_A = - ((B)) +LOG(A)-LOG(A(-1)) $
```

```
[34]: %%Mexplodemodel test show  
><add> dlog(a) = b+c+d+d
```

```
><add> dlog(a) = b+c+d+d
```

```
FRML <ADD> A = A(-1)*EXP(A_A+ (B+C+D+D) ) $
```

```
FRML <CALC_ADD_FACTOR> A_A = - ((B+C+D+D)) +LOG(A)-LOG(A(-1)) $
```

```
[ ]:
```